The Bash Tool: Configuration and Security

The `Bash` tool is the workhorse of ragent. It executes shell commands from the agent's working directory and applies a seven-layer security model.

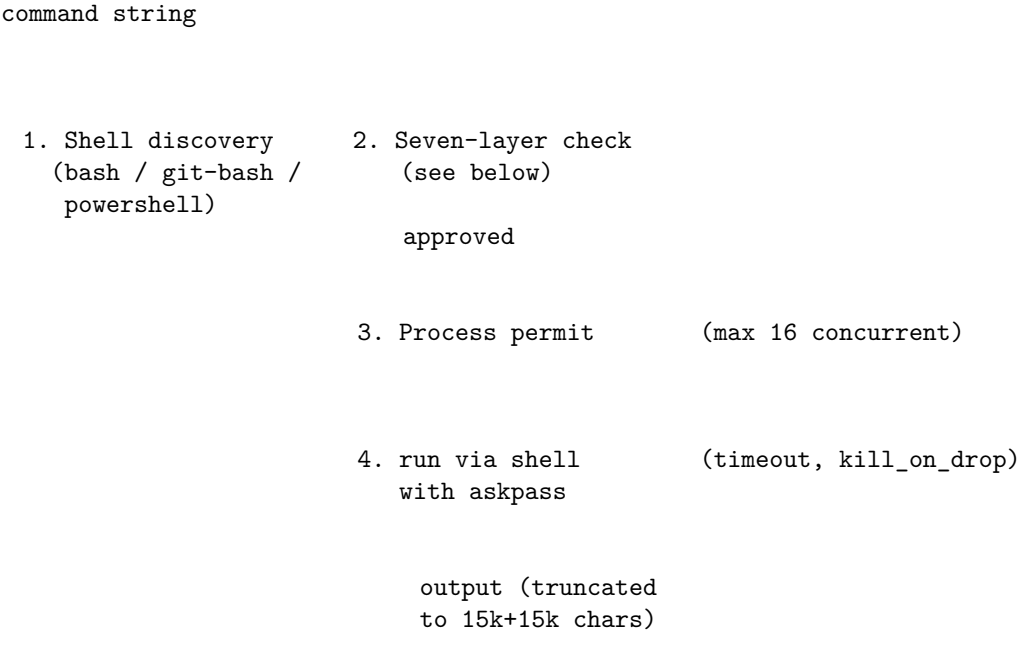
This document covers:

1. Overview
2. Configuration — the `bash` block in `ragent.json`
3. The seven-layer security model
4. The `/bash` slash commands
5. Runtime behaviour — timeout, output truncation, process limits
6. YOLO mode
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Overview

The tool is implemented in `crates/ragent-tools-core/src/bash.rs` (~1750 lines) as the `BashTool` struct. It is a unit struct with no fields of its own — all behaviour is driven by the global `ragent_config` at execution time, populated from your config files.

Every command an agent runs through `Bash` goes through the same pipeline:



Configuration

The Bash tool is configured via the top-level `bash` block in `ragent.json` (global at `~/.config/ragent/ragent.json`, or project at `.ragent/ragent.json`).

The `bash` config block

```
{
  "bash": {
    "allowlist": ["curl", "wget"],
    "denylist": ["git push --force", "systemctl disable"]
  }
}
```

Key	Type	Default	Purpose
`allowlist`	`string[]`	`[]`	Command prefixes exempted from the built-in banned-command check
`denylist`	`string[]`	`[]`	Substring patterns that unconditionally reject a command

The underlying Rust struct (`crates/ragent-config/src/config.rs`):

```rust
pub struct BashConfig {
    /// Command prefixes exempted from the banned-command check.
    pub allowlist: Vec<String>,
    /// Patterns that unconditionally reject a command.
    pub denylist: Vec<String>,
}
```

Merge semantics across global + project config

When both a global and a project config exist, ragent merges them:

- **allowlist** and **denylist** entries are **unioned** (deduplicated). An entry added in either config applies everywhere.

Config flow

The Bash tool does **not** read `Config` directly. At startup (and on `/reload`) ragent copies the merged config into an in-memory snapshot via `bash_lists::load_from_config()` (`crates/ragent-config/src/bash_lists.rs`):

- TUI startup: `App::new` in `crates/ragent-tui/src/app/init.rs`
- `/reload`: the `"reload"` arm in `crates/ragent-tui/src/app/slash.rs`

```rust
pub fn load_from_config() {
    let lists = match crate::config::Config::load() {
        Ok(cfg) => BashLists {
            allowlist: cfg.bash.allowlist,
            denylist: cfg.bash.denylist,
        },
        ...
```

```
    };
}
```

The tool reads from this snapshot at execution time through helpers such as `is_allowlisted()` and `matches_denylist()`.

The seven-layer security model

Every command is validated by seven layers; all seven must pass. The table below summarises them in the order the checks actually execute, full detail follows.

Order	Check	Source	Bypassed by YOLO	Bypassed by allowlist
1	Safe-command whitelist	`SAFE_COMMANDS` const	—	—
2	Banned commands	`BANNED_COMMANDS` const	yes	**yes**
3	Directory-escape	`is_directory_escape_attempt()`	~~()~~	—
4	Syntax validation	`validate_bash_syntax()`	—	—
5	Denied commands & patterns	`DENIED_COMMANDS`, `DENIED_COMMAND_PATTERNS`, `DENIED_PATTERNS` consts	yes	no
6	User denylist	`matches_denylist()`	yes	no
7	Obfuscation detection	`validate_no_obfuscation()`	yes	no

Layer 1 — Safe-command whitelist

A built-in list (`SAFE_COMMANDS` in `crates/ragent-tools-core/src/bash.rs`), mostly read-only but including the file-modifying `mkdir`, `touch`, `cp` and `mv`, is prefix-matched and skips the permission prompt (`validate_shell_command` just logs `"Safe bash command auto-approved"` and still runs every later layer). A command is safe if it **equals** an entry or **starts with the entry followed by a space**, so `ls` matches `ls -la`. A command that contains a shell separator or redirection (`;`, `|`, `&`, a backtick, `$(`, a newline, `>` or `<`) is never treated as safe, because the whitelist only inspects the first token.

```rust
const SAFE_COMMANDS: &[&str] = &[
    // --- File management ---
    "ls", "cd", "pwd", "mkdir", "touch", "cp", "mv",
    // --- File reading & search ---
    "cat", "head", "tail", "grep", "egrep", "fgrep", "rg", // ripgrep
    "wc", "which", "tree", "diff", "file",
    // --- Text formatting (read-only, no argument-driven execution) ---
    "echo", "printf", "sort", "uniq", "cut", "tr", "date",
];
```

> **Note:** The whitelist deliberately contains no version-control, build, package manager or interpreter commands (`git`, `cargo`, `npm`, `python3`, `sed`, `awk`, `find`, `xargs`, …). Those always go through the normal permission gate. `rm` is also excluded so it always requires a permission decision; its destructive forms are caught by `DENIED_PATTERNS` (Layer 5).

Layer 2 — Banned commands

A word-boundary match against `BANNED_COMMANDS` (so `curl` does not match inside `scroll`). These are network / exfiltration and attack tools that are almost never legitimate inside an agent:

```
const BANNED_COMMANDS: &[&str] = &[
    // Network / data-exfiltration tools
    "curl", "wget", "nc", "netcat", "telnet", "axel", "aria2c", "lynx", "w3m",
    // Attack and exploitation tools
    "nmap", "masscan", "nikto", "sqlmap", "hydra", "john", "hashcat",
    "aircrack", "metasploit", "msfconsole", "msfvenom", "burpsuite",
    "ettercap", "arpspoof",
    // Packet capture (enable via YOLO)
    "tcpdump", "wireshark",
];
```

A banned command is rejected **unless**:

- YOLO mode is enabled, **or**
- the command's **first token** is on the user `allowlist` (this is how you re-enable `curl` / `wget` without turning on YOLO).

```
if contains_banned_command(command) {
    if ragent_config::yolo::is_enabled() { /* allow */ }
    else if ragent_config::bash_lists::is_allowlisted(command) { /* allow */ }
    else { bail!("Command rejected: uses banned external tool (curl, wget, nc, ...)"); }
}
```

Layer 3 — Directory-escape prevention

`is_directory_escape_attempt(command, &ctx.working_dir)` rejects commands that `cd` or `pushd` out of the working directory via `..`, `~`, `$HOME`, `${HOME}`, or absolute `/` paths (and, on Windows, `C:\` / `\` drive roots). Absolute paths that stay inside the working directory are allowed after canonicalization, and single-segment slash-prefixed tokens (e.g. `/help`, `/start`) are treated as commands rather than paths. This keeps the agent's filesystem view inside its sandbox. This layer is **never** overridden.

Layer 4 — Syntax validation

`validate_bash_syntax()` runs the command through the shell's own parser with a **1-second timeout** and `kill_on_drop(true)`:

```
bash -n -c "<command>"
```

A non-zero exit produces `Bash syntax error: <stderr>` and the command is rejected. This uses the actual discovered shell (not a hardcoded `sh`) so it works on Git Bash too. Syntax validation is **skipped for PowerShell** (which has its own runtime parser and no `-n` equivalent). This layer is never overridden.

Layer 5 — Denied commands & patterns

Three sub-checks that are **never** bypassed by the allowlist (only by YOLO). Denied commands are checked after heredoc bodies are stripped, so string literals inside a heredoc do not cause false positives.

5a. Denied commands (word-boundary):

```
const DENIED_COMMANDS: &[&str] = &[
    "mkfs", "wipefs",              // Disk / partition destruction
    "insmod",                      // Kernel modifications
    "useradd", "usermod", "groupadd", // User/group manipulation
    "visudo",                      // System configuration
    "grub-install", "efibootmgr",  // Boot/firmware
];
```

5b. Denied command patterns (prefix patterns):

```
const DENIED_COMMAND_PATTERNS: &[&str] = &[
    "sudo ", "sudo\t", "su -", "su root", "doas ", // privilege escalation
    "passwd ",                                     // user/group manipulation
    "crontab -",                                   // system configuration
];
```

5c. Denied substring patterns — the most destructive forms:

```
const DENIED_PATTERNS: &[&str] = &[
    // Destructive filesystem operations
    "rm -rf /", "rm -r -f /", "rm -fr /", "rm -Rf /", "rmdir /",
    // Disk operations
    "dd if=", "shred /dev",
    // Device writes
    "> /dev/sd", "> /dev/nvme", "> /dev/vd",
    // Fork bomb
    ":(){ :|:&};:",
    // Privilege escalation
    "chmod -R 777 /", "chown -R",
    // Network exfiltration of sensitive files
    "curl.*etc/shadow", "wget.*etc/shadow",
    // History / credential file theft
    ".bash_history", ".ssh/id_",
    // Kernel modifications
    "modprobe -r", "sysctl -w",
    // Destructive git operations
    "git push --force", "git push -f ", "git push origin --delete",
    // More destructive patterns
    "rm -rf ~", "rm -rf $HOME", "rm -rf .",
    "chmod 000 /", "chmod -R 000",
    // Data exfiltration via pipes
    "> /dev/tcp", "bash -i >&", "/dev/tcp/", "/dev/udp/",
    "systemctl disable", "systemctl mask", "chattr +i",
];
```

Layer 6 — User denylist

Substring-matches the command against your configured `denylist` patterns. This is where project-specific guardrails go. Bypassed in YOLO mode only.

```
if !ragent_config::yolo::is_enabled()
    && let Some(pattern) = ragent_config::bash_lists::matches_denylist(command)
```

```
{ bail!("matches user-defined deny pattern ..."); }
```

> **Important asymmetry:** an allowlist entry only exempts a command from **Layer 2 (banned commands)**. It does **not** exempt it from the denylist (Layer 6) or the denied-pattern checks (Layer 5). That is by design.

Layer 7 — Obfuscation detection

`validate_no_obfuscation()` rejects clearly-obfuscated payloads:

- `base64 ... | bash` / `| sh`
- `python` / `perl` using `exec(` / `eval(`
- `$'\xNN` hex-escape sequences
- `eval` combined with `$(...)` command substitution

Bypassed in YOLO mode only.

The `/bash` slash commands

The TUI exposes interactive management of the allowlist/denylist via `/bash`:

```
/bash add allow <cmd>           # add a command prefix to the allowlist
/bash add deny <pattern>        # add a deny pattern
/bash remove allow <cmd>        # remove an allowlist entry
/bash remove deny <pattern>     # remove a deny pattern
/bash show                      # display current allowlist / denylist
/bash help
```

Each `add`/`remove` accepts an optional `--global` flag to target the global config (`~/.config/ragent/ragent.json`) instead of the project config (`.ragent/ragent.json`). Changes persist to disk immediately and update the in-memory snapshot used by the tool.

Runtime behaviour

Timeout

The default timeout for a shell command is **120 seconds** (`DEFAULT_TIMEOUT_SECS`). It can be overridden per-call via the `timeout` parameter on the tool. On timeout the tool returns `"Command timed out after N seconds"` with `timed_out: true` metadata.

Process limits

Before spawning, the tool acquires a process permit from a semaphore that caps **concurrent shell processes at 16** (`crates/ragent-types/src/resource.rs`). Combined with a separate cap of 5 concurrent tools, this prevents an agent from forking runaway process trees.

`kill_on_drop` and process groups

Every process builder sets `kill_on_drop(true)` **and** `process_group(0)`. `kill_on_drop(true)` kills the *direct child* if the command's future is dropped; it does not reach grandchildren. On **timeout**,

ragent additionally SIGKILLs the entire process group via `kill_process_group` (`libc::killpg`), so descendant processes (e.g. a deadlocked `cargo test` child) are reaped too. Without the timeout path, a plain dropped future kills the direct child only. This closed a real bug fixed in v1.0.60 — repeated bash timeouts previously accumulated orphaned processes. The flags are present at every spawn site (syntax check, all three execute branches, and all background-shell paths).

Post-exit drain and capture cap

After a command exits, the tool drains remaining pipe output for up to a bounded window (`POST_EXIT_DRAIN_TIMEOUT_SECS` = 5 s) so trailing lines are captured without hanging, and per-stream in-memory capture is capped at `MAX_CAPTURE_BYTES` = 2 MiB to bound memory on chatty processes.

Output truncation

Output is truncated to the **first 15,000 + last 15,000 characters** (`MAX_OUTPUT` = 31,000) with UTF-8 boundary alignment and an `… lines omitted …` separator, so very long command output never blows up the agent context.

Persistent shell state & working directory

The tool maintains a per-session state file that records the current working directory. Commands can `cd` within the sandbox and ragent tracks the change (publishing `Event::ShellCwdChanged` on the event bus) so subsequent commands run from the updated directory.

Sudo askpass broker

For the rare legitimate `sudo` case, ragent starts an `AskPassBroker` that routes the `SUDO_ASKPASS` prompt through the interactive question dialog instead of hanging on a terminal password prompt.

YOLO mode

YOLO mode (`crates/ragent-config/src/yolo.rs`) is an escape hatch for trusted, single-user environments. It bypasses **Layers 2, 5, 6, and 7** (banned commands, denied commands/patterns, user denylist, and obfuscation detection). Layers 1, 3, and 4 (safe-command whitelist, directory-escape and syntax validation) remain active — YOLO does **not** let the agent escape its working directory or run syntactically-invalid commands.

Enable it with `/yolo`, `Alt+Y`, or by setting `"yolo": true` in config. Once enabled in either the global or project config it stays enabled (OR merge semantics).

Platform notes

- **PowerShell:** syntax pre-validation (Layer 4) is skipped; the runtime parser inside the wrapper handles it. All other layers apply.

Reference

Files

Path	Role
`crates/ragent-tools-core/src/bash.rs`	The `BashTool` implementation
`crates/ragent-tools-core/src/bg.rs`	Background shell integration
`crates/ragent-config/src/config.rs`	`BashConfig` struct and defaults
`crates/ragent-config/src/bash_lists.rs`	Runtime allowlist/denylist snapshot
`crates/ragent-config/src/yolo.rs`	YOLO mode
`crates/ragent-tui/src/app/slash.rs`	`/bash` slash commands
`crates/ragent-tui/src/app/init.rs`	`load_from_config()` at startup
`crates/ragent-types/src/resource.rs`	Process/tool semaphores (16 processes, 5 tools)

Config JSON keys (summary)

Key	Type	Default	Meaning
`bash.allowlist`	`string[]`	`[]`	Command prefixes that bypass the banned-command check
`bash.denylist`	`string[]`	`[]`	Substring patterns that always reject a command
`yolo`	`bool`	`false`	Master switch that bypasses security layers 2, 5, 6, 7

Examples

Re-enable `curl` (without YOLO) but keep it out of `rm` destructive forms:

```
{
  "bash": {
    "allowlist": ["curl", "wget"],
    "denylist": ["rm -rf /", "git push --force"]
  }
}
```

This document describes the Bash tool as implemented in ragent v1.0.81.